ECON9104C Topics in Econometrics

Exam

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March 13, 2025

1 Introduction

The COVID-19 pandemic introduced a major disruption to cross-border labor mobility. In many countries, temporary border closures restricted the inflow of non-resident workers who typically commute across national borders for employment. These restrictions can be interpreted as an increase in mobility frictions affecting firms that rely on cross-border labor supply.

In this project, I aim to use a static random utility model to study how such mobility frictions interact with labor market conditions and affect firms' hiring decisions. In particular, firms may choose between hiring resident workers and non-resident workers. When cross-border mobility becomes more difficult, firms may need to substitute toward domestic labor or adjust their hiring strategies. Understanding how firms respond to such shocks is important for analyzing the effects of mobility barriers on labor market outcomes.

To capture these mechanisms, I develop a structural model of firms' hiring decisions in which firms choose between alternative worker types within a probabilistic discrete-choice framework. The model allows hiring probabilities to depend on wages, labor market tightness, firm characteristics, and mobility frictions affecting non-resident workers.

The remainder of the hand-in proceeds as follows. Section 2 introduces the Beveridge curve and the concept of labor market tightness. Section 3 presents the structural model and its empirical specification. Section 4 describes the simulation and estimation procedure, and Section 5 presents the results from the simulated data and Monte Carlo experiment.

2 The Beveridge (UV) Curve

The Beveridge curve (or UV curve) describes the empirical negative relationship between unemployment and job vacancies (Beveridge, 1944). It provides a reduced-form characterization of labor market tightness and matching frictions, such that periods with high unemployment are associated with few vacancies (slack labor markets), while periods with many vacancies coincide with low unemployment (tight labor markets). Movements along the curve typically reflect cyclical fluctuations in labor demand, whereas outward or inward shifts indicate changes in matching efficiency or structural frictions such as search

costs, skill mismatch, or mobility barriers.

The labor market tightness is represented by a vacancy-unemployment ratio that summarizes how difficult it is for firms to fill open positions, defined as

$$\theta_t \equiv \frac{v_t}{u_t}, \tag{1}$$

where v_t denotes the vacancy rate and u_t the unemployment rate at time t. Higher values of θ_t correspond to tighter labor markets and higher expected recruiting costs for firms. In this project, I intend to treat tightness as an observable state variable.

3 Structural Model

To study how labor market frictions affect firms' hiring behavior, I model labor demand using a static probabilistic choice framework. There are two sources of labor: either resident workers (R) or non-resident workers (NR). Non-resident workers are those who have their workplaces in Norway, although they do not live in Norway permanently. For example, these could be commuters who travel daily across the border from Sweden, or workers from East Europe that are flown in to Norway to work at a firm for a short period of time before returning to their home country.

Each firm picks their source of labor between resident workers (R) and non-resident workers (NR). Let $n = 1, \ldots, N$ index firms and $i \in \{R, NR\}$ index worker types.

3.1 Utility Specification

The profit obtained by firm n from hiring worker type i at time t is

$$U_{ni,t} = V_{ni,t} + \varepsilon_{ni,t}, \tag{2}$$

where $V_{ni,t}$ is the systematic component of profits and $\varepsilon_{ni,t}$ captures unobserved match-specific productivity or hiring costs. The systematic component represents expected profits from hiring a worker, determined by worker productivity, wages, and recruiting costs.

I parameterise the systematic component as

$$V_{ni,t} = \beta_p \ln p_i - \beta_w \ln w_{i,t} - \beta_f \ln \theta_t - \beta_b \mathbf{1}\{i = NR\} B_t, \tag{3}$$

where

- p_i denotes the productivity of worker type i,
- $w_{i,t}$ is the wage of worker type i at time t,
- θ_t captures labor market tightness and recruiting frictions,
- B_t represents an additional regime affecting non-resident workers.

Higher wages or tighter labor markets reduce expected profits from hiring, while higher productivity increases them. Higher wages increase the cost of employing a worker, whereas tighter labor markets raise recruiting costs by making workers harder to find. Higher productivity, in contrast, increases the output generated by the match and therefore raises the profitability of hiring.

Assuming that the shocks $\varepsilon_{ni,t}$ are independently and identically distributed according to a Type I extreme value distribution, the probability that firm n hires worker type i follows the multinomial logit form

$$P_{ni,t} = \frac{\exp(V_{ni,t})}{\sum_{j \in \{R,NR\}} \exp(V_{nj,t})}. \tag{4}$$

3.2 Decision Rule

The firm chooses the worker type that yields the highest profits. Formally, firm n hires worker type i at time t if

$$U_{ni,t} \geq U_{nj,t} \quad \forall j \neq i. \tag{5}$$

With two alternatives, resident workers and non-resident workers, the firm hires a non-resident worker whenever

$$U_{n,NR,t} - U_{n,R,t} \geq 0. \tag{6}$$

The utility difference can therefore be written as

$$\Delta U_{n,t} = \Delta V_{n,t} + \Delta \varepsilon_{n,t}, \tag{7}$$

where

$$\Delta V_{n,t} = \beta_p \ln \left(\frac{p_{NR}}{p_R} \right) - \beta_w \ln \left(\frac{w_{NR,t}}{w_{R,t}} \right) - \beta_b B_t. \tag{8}$$

Under the assumption that the shocks follow a Type I extreme value distribution, the probability that firm n hires a non-resident worker takes the binary logit form

$$P_{n,NR,t} = \frac{\exp(\Delta V_{n,t})}{1 + \exp(\Delta V_{n,t})}. \tag{9}$$

3.3 Empirical Specification

To estimate the model, I parameterise the systematic profit difference as

$$\Delta V_{n,t} = \alpha + \beta_w \ln w_{ratio,t} + \beta_f \ln \theta_t + \beta_b B_t + \beta_x X_n, \tag{10}$$

where

- $\ln w_{ratio,t} = \ln \left(\frac{w_{NR,t}}{w_{R,t}} \right)$ captures the relative wage cost of hiring non-resident workers,
- $\ln \theta_t$ denotes labor market tightness,
- B_t is a regime variable affecting non-resident workers,
- X_n is a firm-specific characteristic affecting hiring decisions.

The variable $\ln w_{ratio,t}$ captures the relative wage cost of hiring non-resident workers compared to resident workers. Because firms choose between alternative worker types, the relative wage difference determines which type is more profitable to hire.

Labor market tightness θ_t is defined as the vacancy–unemployment ratio and summarizes how difficult it is for firms to fill vacancies. In the context of the Beveridge curve, movements in θ_t reflect changes in overall labor market conditions. Higher values correspond to tighter labor markets where firms face greater difficulty recruiting workers.

The variable B_t captures a regime affecting non-resident workers, such as a border closure that affects thair mobility. Changes in this variable represent shifts in the institutional environment that may alter firms' ability or incentives to hire workers from abroad.

Finally, X_n represents firm-specific characteristics that may affect hiring behavior. These characteristics capture heterogeneity across firms in their demand for different types of labor.

The probability that firm n hires a non-resident worker is therefore

$$P_{n,NR,t} = \frac{\exp\left(\alpha + \beta_w \ln w_{ratio,t} + \beta_f \ln \theta_t + \beta_b B_t + \beta_x X_n\right)}{1 + \exp\left(\alpha + \beta_w \ln w_{ratio,t} + \beta_f \ln \theta_t + \beta_b B_t + \beta_x X_n\right)}. \tag{11}$$

3.4 Identification

The parameters are identified from variation in relative wages, labor market conditions, policy regimes, and firm characteristics.

- β_w is identified from variation in the wage differential between resident and non-resident workers. Changes in the relative wage affect the profitability of hiring different worker types and therefore influence firms' hiring probabilities.
- β_f is identified from variation in labor market tightness θ_t. Changes in the vacancy–unemployment ratio capture fluctuations in overall labor market conditions and recruiting difficulty, which affect firms' incentives to hire different types of workers.
- β_b is identified from changes in the regime variable B_t affecting non-resident workers. In the context of the model, this variable captures shifts in mobility frictions, such as border closures during the COVID-19 pandemic.
- β_x is identified from cross-firm variation in observable characteristics X_n, which capture heterogeneity in firms' demand for different types of labor.

3.5 Estimation

The model parameters are estimated using maximum likelihood. Let $y_{n,t}$ denote the observed hiring decision, where $y_{n,t} = 1$ if firm n hires a non-resident worker and $y_{n,t} = 0$ otherwise. The likelihood contribution of observation (n, t) is

$$L_{n,t} = P_{n,NR,t}^{y_{n,t}}(1 - P_{n,NR,t})^{1-y_{n,t}}. \tag{12}$$

The sample log-likelihood is therefore

$$\mathcal{L}(\theta) = \sum_{n,t} \left[y_{n,t} \ln P_{n,NR,t} + (1 - y_{n,t}) \ln(1 - P_{n,NR,t}) \right], \tag{13}$$

where $\theta = (\alpha, \beta_w, \beta_f, \beta_b, \beta_x)$ denotes the vector of model parameters.

Then parameter estimates are obtained by maximizing the log-likelihood with respect to θ.

4 Simulation and Implementation

This section describes how the structural model is implemented in Python and how simulated data are generated, estimated, and then used for the counterfactual analysis.

4.1 Implementation of the Choice Probability

The choice probability implied by the logit model is implemented using a logistic transformation:

```
def sigmoid(z):
    z = np.clip(z, -35, 35)
    return 1.0 / (1.0 + np.exp(-z))
```

This function computes the logistic transformation

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \tag{14}$$

which maps any real value into the interval $(0, 1)$. In the model, this transformation converts the systematic profit difference $\Delta V_{n,t}$ into the probability that firm n hires a non-resident worker.

The systematic component of the model is implemented in the function

```
def solve_model(params, df):
```

where the parameter vector is

$$(\alpha, \beta_w, \beta_f, \beta_b, \beta_x). \tag{15}$$

The function computes the systematic profit difference

$$\Delta V_{n,t} = \alpha + \beta_w \ln w_{ratio,t} + \beta_f \ln \theta_t + \beta_b B_t + \beta_x X_n, \tag{16}$$

and then transforms it into the predicted hiring probability

$$p_{n,t} = \frac{\exp(\Delta V_{n,t})}{1 + \exp(\Delta V_{n,t})}. \tag{17}$$

The function returns the predicted probability for each observation in the dataset.

4.2 Simulating Synthetic Panel Data

To evaluate the estimator, artificial panel data are generated using the structural model. The simulation function is

```
def simulate_data(N=2000, T=24, true_params=None, seed=221):
```

which returns a dataset containing the variables `ID`, `t`, `X`, `ln_theta`, `B`, `ln_w_ratio`, and the simulated hiring decision `y`.

If no parameter vector is supplied, the simulation uses the default parameters

```
true_params = (-0.2, -1.0, 0.8, 0.6, 0.4)
```

corresponding to $(\alpha, \beta_w, \beta_f, \beta_b, \beta_x)$. Because these parameters are known, the simulation can be used to evaluate whether the estimation procedure is able to recover them.

The simulation generates firm-level heterogeneity through a firm characteristic X_n and constructs time-varying labor market conditions. Labor market tightness follows a cyclical process with additional random variation, while the regime variable B_t introduces a policy shift affecting non-resident workers. Relative wages are generated to be correlated with labor market tightness.

Firm-level and time-level variables are combined to produce a balanced panel with $N \times T$ observations. Given the simulated explanatory variables, hiring probabilities are computed using the structural model and the observed hiring decision is generated as

$$y_{n,t} \sim \text{Bernoulli}(p_{n,t}). \tag{18}$$

The resulting dataset follows the structure implied by the theoretical model and can therefore be used to evaluate the estimator in a controlled setting.

4.3 Maximum Likelihood Estimation

The parameters are estimated by maximum likelihood. The negative log-likelihood function is defined as

```
def neg_loglik(params, df):
```

For a given parameter vector, the model computes the predicted hiring probability $p_{n,t}$. Since the observed hiring decision is binary, the likelihood contribution of observation (n, t) is

$$L_{n,t} = p_{n,t}^{y_{n,t}} (1 - p_{n,t})^{1-y_{n,t}}. \tag{19}$$

The sample log-likelihood is therefore

$$\mathcal{L}(\theta) = \sum_{n,t} \left[y_{n,t} \ln p_{n,t} + (1 - y_{n,t}) \ln(1 - p_{n,t}) \right]. \tag{20}$$

The estimator is obtained by minimizing the negative log-likelihood using the BFGS optimization algorithm implemented in `scipy.optimize`. BFGS is a quasi-Newton method that iteratively approximates the Hessian matrix and is widely used for maximum likelihood estimation in nonlinear models.

5 Results

This section presents the results obtained from the simulated data. I first report the estimation results from a single simulated dataset and compare the estimated parameters with the values used in the data-generating process. I then use the estimated model to illustrate several counterfactual scenarios. Lastly I run a Monte Carlo experiment.

5.1 Estimation Results from a Single Simulation

The model is first estimated using a simulated dataset with $N = 3000$ firms observed over $T = 24$ time periods. The parameter values used in the data-generating process are

$$(\alpha, \beta_w, \beta_f, \beta_b, \beta_x) = (-0.2, -1.0, 0.8, -0.6, 0.4). \tag{21}$$

The corresponding parameter estimates obtained from the maximum likelihood procedure are reported in Table 1.

Parameter	True value	Estimated value
α	-0.20	-0.208
β_w	-1.00	-1.041
β_f	0.80	0.807
β_b	0.60	-0.582
β_x	0.40	0.387

Table 1: Estimated parameters from a single simulated dataset

The estimated coefficients are close to the parameter values used in the simulation. This suggests that the maximum likelihood estimator is able to recover the structural parameters reasonably well in samples of this size.

The estimated coefficients also have economically intuitive signs. The negative coefficient on the wage ratio implies that higher relative wages for non-resident workers reduce the probability that firms hire them. In contrast, the positive coefficient on labor market tightness indicates that firms are more likely to hire non-resident workers when the labor market becomes tighter.

5.2 Counterfactual Results

Using the estimated parameters, the model can be used to evaluate simple counterfactual scenarios. Table 2 reports the predicted average probability that a firm hires a non-resident worker under several alternative environments.

Scenario	Predicted hiring probability
Baseline	0.386
Tightness +10%	0.404
Open border ($B = 0$)	0.445
Closed border ($B = 1$)	0.322

Table 2: Counterfactual predicted hiring probabilities

Increasing labor market tightness by 10% raises the predicted probability that a firm hires a non-resident worker from 0.386 to 0.404. This reflects the positive relationship between labor market tightness and firms' incentives to recruit workers from outside the local labor market.

Changing the regime variable also leads to substantial differences in predicted hiring behavior. When the border regime corresponds to an open-border environment ($B = 0$), the predicted hiring probability increases to 0.445. In contrast, when the regime represents a closed-border environment ($B = 1$), the predicted probability falls to 0.322. These counterfactual exercises illustrate how restrictions on cross-border mobility can reduce firms' reliance on non-resident workers.

5.3 Monte Carlo Design

To evaluate the finite-sample performance of the estimator, I conduct a Monte Carlo experiment. The full simulation and estimation procedure is repeated $R = 200$ times. In

each replication, a new dataset is generated from the same structural model using different random draws.

For each simulated dataset, the model parameters are estimated by maximum likelihood and the resulting estimates are stored. This produces a distribution of parameter estimates across repeated samples, which can be used to assess whether the estimator recovers the true structural parameters on average.

5.4 Monte Carlo Results

Table 3 reports the mean parameter estimates across the 200 Monte Carlo replications together with the true parameter values used in the data-generating process.

Parameter	True value	Mean estimate
α	-0.20	-0.200
β_w	-1.00	-1.003
β_f	0.80	0.800
β_b	-0.60	-0.599
β_x	0.40	0.401

Table 3: Monte Carlo results (200 replications)

The mean estimated coefficients are very close to the true parameter values used in the simulation. For example, the true wage coefficient is -1.0 while the average estimate across replications is approximately -1.003. Similarly, the true tightness parameter is 0.8 and the average estimate is approximately 0.800.

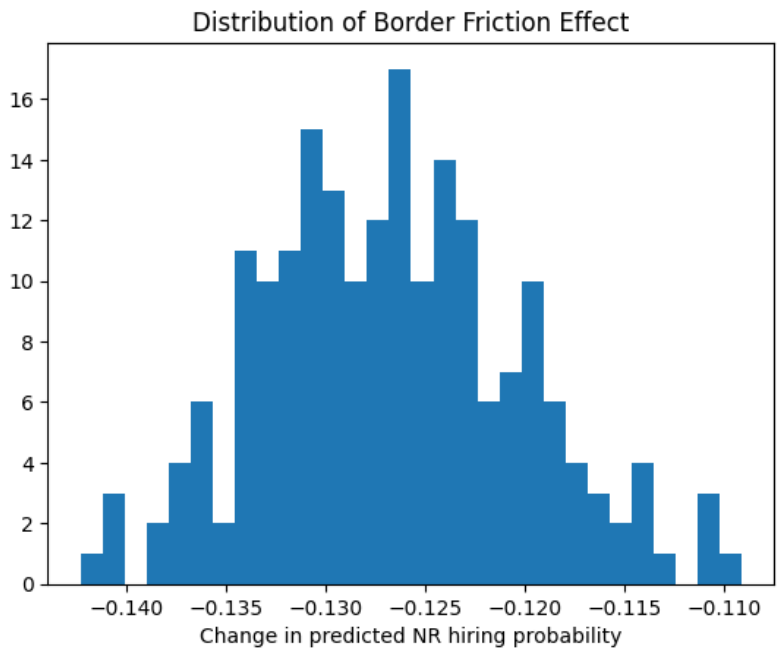

Figure 1: Sampling distribution of the estimator $\hat{\beta}_f$ across 200 Monte Carlo replications.

Figure 1 illustrates the sampling distribution of the estimator $\hat{\beta}_f$ across the Monte Carlo replications. The distribution is centered close to the true parameter value, which is consistent with the small bias reported in Table 3.

Overall, the Monte Carlo experiment suggests that the maximum likelihood estimator performs well in samples of this size and is able to recover the structural parameters of the model with very little bias.

6 Conclusion

This hand-in develops a simple structural model of firms' hiring decisions between resident and non-resident workers. The model combines a random utility framework with observable labor market conditions, allowing hiring probabilities to depend on wages, labor market tightness, firm characteristics, and a regime variable affecting cross-border mobility.

Using simulated panel data, the parameters of the model are estimated by maximum likelihood. The results from a single simulated dataset show that the estimator is able to recover the true parameter values with reasonable accuracy. A Monte Carlo experiment with 200 replications further indicates that the estimator performs well in finite samples

and produces estimates that are close to the parameters used in the data-generating process.

The estimated model can also be used to evaluate counterfactual scenarios. Changes in labor market tightness and the border regime generate meaningful differences in predicted hiring probabilities, illustrating how the framework can be used to study the effects of labor market conditions and mobility restrictions on firms' hiring behavior.

Overall, the exercise demonstrates how a structural discrete-choice model can be implemented, estimated, and used for simple policy counterfactuals using simulated data.

References

Beveridge, W. (1944). *Full employment in a free society.* George Allen & Unwin.